

# “In esports teamwork ranks way higher than anything else”: An exploration of esports training perceptions

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## ABSTRACT

Esports has been a growing area of study in recent years. We conducted two qualitatively focused survey studies to further understand the landscape of esports training. Specifically, via research questions related to coaching and training programs, types of training, gaps in training routines, how different people perceive training, and how important various training categories are perceived. The first was a pilot survey exploring existing esports training, the second was a more in-depth exploration of esports training and how fans and specialists perceive the field. We found that fans and specialists generally agreed on many aspects of esports training. For example, teamwork was widely agreed upon as a crucial aspect of training. However, specialists seemed more focused on mental and physical health than fans. Further, three general domains emerged as overarching trends worthy of focus: technical skills, teamwork, and health support (both physical and mental). These domains align well with prior researchers' findings. Our results also support the notion that greater organization and focus should be placed on teamwork and mental health support, as those areas are perceived to be under served. We also found a strong undercurrent of a lack of access (primarily due to physical location) and a desire for more guidance in the training process. Therefore, future efforts should be made to make esports training resources, particularly mental health resources, more well-known and accessible.

## 1. Introduction

Esports professionals do not rely solely on natural talent. Like other performers and athletes, professional esports players must train to keep their skills sharp. We conducted two survey studies focused on exploratory qualitative data to explore what esports specialists and fans think about existing training and what should be improved. Since this is such a broad notion, we began with a 2021 special interest group's findings. They identified several areas of esports research that should be expanded on [1]. These areas included streamer and spectator experience, health (physical/cognitive/mental) in esports, performance improvement in esports, modeling and strategy identification of esports behaviors, team formation and interpersonal dynamics in esports, plus some additional areas [1]. The current study adds to the collective knowledge of the domains mentioned and serves as a launching pad for future projects.

More specifically, the current research is intended to assess the training procedures that esports industry professionals use and those who wish to break into the industry. We refer to these people throughout as “specialists” to cover professionals, aspiring professionals, coaches, managers, support staff, players on nonprofessional teams

(like school teams), and retired professionals. We also wish to assess what training areas are perceived as the most and least important, and where gaps are perceived. Further, we aim to compare fan perceptions with specialists to better understand how well the overarching fandom understands the professional player experience, particularly since at least some professionals are emerging from the pool of fans [2]. Two survey studies are detailed in this paper. The first was a 28-question pilot survey, where we recruited participants via personal networks and social media. This study was used to refine our survey for the second study. The second was a 16 to 22-question survey wherein the service *Game Tester* was used for recruitment [3]. The number of questions participants received for both studies depended on whether the participants were esports specialists or fans.

The following paper discusses the overall design, related work, overall methodology, and research questions. Then, the methods for Studies 1 and 2. Afterward, we discuss the results for Studies 1 and 2. The paper ends with a discussion of both studies. One recent paper by Formosa et al. [4] notes that according to the AP style guide, “esports” is correct. Thus, we use it throughout. However, we did not alter the formatting in the direct quotes of the participants [4]. For our

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purposes, we define esports as the play and spectating of competitive online games within an established competitive environment with an agreed-upon rule set [5].

## 2. Related work

Esports requires interaction with video games, so we began with a question: what factors can impact game-play environments? One conceptualization of such factors involves breaking factors into two categories [6,7]. In one paper [6], the two categories were out-of-game and in-game. Out-of-game included items like cultural differences and gender, while in-game included items like game mechanics and tactical awareness. Another paper breaks factors into similar categories: virtual environment experience and in-person social outcomes [7]. Other researchers have also found that brain activation is different depending on gamer skill, with one study finding that people who were more skilled in games showed activation of different brain areas than less skilled participants [8]. Another publication reported that emotions, cognition, and physiological arousal seem to be interconnected with esports [9]. Thus, when we discuss the factors involved with esports, there are a multitude of possible directions. In response, we organize the following section by discussing cognition, then transition to the biopsychosocial model (a health model combining the influences of biological, psychological, and social health) and its connection to esports generally [10–14]. Afterward, we narrow our focus specifically to prior research on esports training.

### 2.1. Cognition

There is a logical connection between cognition and video games/esports, as cognition involves the perception and processing of stimulation. An early meta-analysis studying cognition and video games found that video game play led to improved information processing in both quasi-experiments and true experiments [15]. Another meta-analysis looked at action video games specifically [16]. This analysis found that action video games have at least some connection to cognitive skills [16]. The connection between action video games and cognition is also present in a 2018 review [17]. Other texts in this area found enhanced cognitive function and superior performance on cognition tasks among gamers vs non-gamers [18,19].

Further, a review reported that nutrition and dietary supplements should be considered part of keeping the cognition of esports players at optimal levels [20]. Nutrition is also associated with health, and a poor diet has been linked to esports, and thus, we transition into discussing the relationship between health and esports [21].

### 2.2. Esports and health

One of the most commonly used models for human health is the biopsychosocial model, which states that a person's overall health is determined by biological, psychological, and social factors [13,14]. Still, this model is very general, so several specific models of esports health have been proposed recently [10–12].

For example, the Model of Health in Esports included physical health, psychological wellness, emotional support, and daily routines [10]. Per their research, all of these subareas seem to be interconnected [10]. Similar factors are found elsewhere, as another review found three primary domains relating to esports and health [11]. These three categories included esports and physical issues (like eye strain and back problems), esports and psychological issues (like higher levels of stress), and esports and addiction [11]. The last review outlined a “three-point framework” on general esports health, though this model does not align as well with the biopsychosocial model [12]. The three points of this framework included “(1) awareness and management of common musculoskeletal and health hazards, (2) opportunities for

health promotion, and (3) recommendations for performance optimization”. They also recommended that esports management include access to a medical professional or team of medical professionals [12]. These models have a lot in common with the biopsychosocial model, but are not the same.

#### 2.2.1. Esports and physical health

Regarding physiological responses, one paper [22] found that esports players experienced increased systolic blood pressure and decreased accuracy and inhibitory processes. First-person shooter games led to a higher peak heart rate than Multiplayer Online Battle Arena (MOBA) games [22].

Another review by Chan et al. found several trends in esports player health, including increased sedentary behavior, higher Body Mass Index (BMI), and poor nutrition. The findings of Chan et al. [21] do not go unchallenged, however. Shulze et al. [23] found in a biopsychosocial model-focused review that BMI and sedentary behaviors were no higher risk in the esports population than the regular population [21,23].

Sleep is an essential health domain [10,11,20,24–26]. Interestingly, the Chan et al. review did not find that esports players had difficulties with sleep quality [21], though other researchers have found a link between sleep health and esports players [24,25]. One paper notes that esports professionals likely deal with many sleep disruptions, which might impact the quality of their sleep. Examples include caffeine, late nights, and air travel [24]. Another study supports these ideas [25]. They found that esports professional athletes had “delayed sleep patterns and have prolonged wake after sleep onset. These sleep patterns may be associated with mood (depression) and training time” [25]. So many of these factors are not as clearly cut as they might seem at first.

#### 2.2.2. Esports and psychological health

Esports also has links to psychological factors like self-esteem [27]. Another review of eight studies examining the psychological aspects of pursuing an esports career found several factors important to esports players [28]. These factors include that players need to have game knowledge, think strategically and be decisive, be motivated and avoid rumination on losses, be able to separate daily life from the game, avoid distraction and stay focused, use constructive coping mechanisms when dealing with harassment, keep a growth mindset, and do physical/mental warm-ups before performances [28].

Other researchers are interested in psychological and physiological stress in esports [10,26,29–34]. One review paper found that playing esports games does not have a clear-cut impact on psychophysiological stress responses [29]. Meanwhile, another paper's results suggested that esports athletes' coping skills could benefit from access to the psychological interventions designed for traditional athletes [31]. Finally, another paper found that cortisol levels increased between pre- and post-game in esports players [32]. They also found that winners and losers generally differed very little, but winners showed higher levels of anxiety. Thus, this anxiety may be beneficial to performance [32].

The paper proposing the Model of Health in Esports found that their professional and amateur participants heavily emphasized psychological issues like burnout and stress as problems [10]. In one study, a group of esports players identified two primary groups of stressors: internal stressors (like communication, team discipline, and individual issues like work/life balance) and external stressors (like public criticism and logistics) [33]. Players reported several coping strategies, which the authors of the paper organized into five categories: emotion-focused coping (like comfort eating), problem-focused coping (like communicating with teammates about encountered issues), approach coping (like working on overall team cohesiveness), avoidance coping (like avoiding forums where they would be given feedback), and appraisal coping (via things like reframing the situation to seem less bad) [33]. Another study completed a thematic analysis of interviews and found general themes important to esports player coping skills: life balance (essentially separation between the game and life), social

support, and good sleep [26]. Finally, another paper oriented around coping and burnout found that better resilience factors could improve coping skills and strategies, and the authors recommend future studies on resilience training for esports players [35].

### 2.2.3. Esports and social health

In the biopsychosocial model, a person's social situation is associated with better health outcomes, and esports play has been linked to positive social health outcomes like improved teamwork and communication skills [27]. One review reported a link between social (or multiplayer) video game play and general well-being [36]. Other papers emphasize the importance of team dynamics and communication relevant to player well-being and success [7,37], with one case study finding that teams who communicated more were more successful [37].

Social support can take many forms, and one study reported four discrete themes for social support in esports [38]. These four themes were emotional support, informational support (support when learning new information), esteem support (support oriented around self-development and improvement), and instrumental support (often referred to as "teamwork" by the sample). In this study, all participants, whether professionals or amateurs, reported that some social support came from their teammates over time [38].

## 2.3. Esports training

The area of Esports Training specifically also yields a lot of interesting findings relevant to the current study.

### 2.3.1. Physical training

For example, some of the research in the domain of physical exercise seems to conflict. One review found that esports players did seem to engage in regular physical activity [39]. In another survey and interview paper, researchers found that most esports participants believed that their performance improved with physical conditioning and reported that physical training took up one hour of their daily average, totaling a little over five training hours per day [40]. However, another paper reported that forty percent of collegiate esports players did not participate in any physical exercise [41]. There was also a review that cited the two previously mentioned papers [40–42]. They recommended further exploration into a concept called "Exergames", which are video games controlled via physical body movements [42]. A paper published after the aforementioned review also found that among the professional esports players they interviewed, only three reported exercise as part of their routine, while the amateur sample reported an average of 1.5 h daily for about five days each week [10].

### 2.3.2. Training time/routines

Two previously mentioned studies reported esports training times. Lee et al. [25] found that different teams spent different amounts of time training [25]. The esports professionals they worked with spent an average of 9.21 h training per day across three countries: South Korea (trained for an average of 13.38), Australia (trained for an average of 4.75), and the United States (trained for an average of 6.10 h) [25]. The other paper reported that their sample of amateur esports players practiced for an average of 5.03 h per day [10]. Thus, daily training time does seem to vary substantially across different teams. Another study also found that training was reported as accounting for 39 percent of participants' total time playing games [43].

Training routines were emphasized in the Model of Health in Esports [10]. While maintaining a training routine individually and without any program is possible, players would likely have more decisive direction if a coach is present or if they were using purpose-built programs. In one study of esports gamers, of the participants who said they had taken part in a tournament or league, more than half said they used structured training programs, and around thirty percent said they had a coach [44]. Thus, many esports players are training with direction and feedback.

### 2.3.3. Teamwork

Teamwork and team coordination/formation in esports have also been studied in the context of training [7,39,45,46]. Team communication was listed as important, as was coordinating differences in personality and game-play preferences [46]. Another study found that "Team cognition" (the ability to process and share information, also can be understood as "being on the same page") was improved via preparation, communication, and a "positive social atmosphere" [45].

### 2.3.4. Training priorities

One aspect of the training we were interested in was reviewing game-play to improve performance. One paper we read to this end found that different game genres had different desires and priorities when reviewing their game-play [47]. In another study by Nagorsky and Wiemeyer, participants rated different dimensions of the *Integrative Model* differently in importance, and these differences are likely due to the different esports games [43]. The *Integrative Model* lists 19 dimensions believed to be relevant to esports players [43].

## 3. Methodology

Section 2 shows that although esports training is actively being studied, more data is still needed for a more complete picture. To that end, the current study is meant to ascertain further information on esports training procedures. For example, what is already in use, and where the gaps are in existing procedures. Also, how do experts perceive the field differently from fans and noncompetitive gamers? We also want to see if our findings support prior research. The related work shows that these dimensions are somewhat nebulous, with some clear trends (like mental health and coping skills being important) and other areas being less clear (like physical activity).

To pursue this end, we designed two qualitative data-focused survey studies. Study 1 is more exploratory than Study 2. The differences between the two studies are articulated in Fig. 1.

The strongest influence on the current study was Nagorsky and Wiemeyer's 2020 paper [43]. They asked participants to rate the importance of several skill dimensions on a Likert scale. We repeat this using their *Integrative Model* but include a forced-choice rating portion to force participants to choose which items are the most/least important, as they can rate things as equally important via Likert scales [43].

We hope to approach some answers to our Research Questions (RQs) with both studies and prior research taken together. The RQs are as follows:

- RQ 1: Are players training by following a coach/program? What do these programs consist of? Are they interested in a program/coaching? How does the coaching/program differ from independent training?
- RQ 2: What kind of training is done outside of just playing the esports game (like spectating, physical/mental training, etc.)?
- RQ 3: Where do participants perceive gaps/problems with current training regimes?
- RQ 4: How are esports fans and specialists perceiving training? Do fans and specialists agree?
- RQ 5: How do our participants rank the items in the *Integrative Model* presented in Nagorsky and Wiemeyer [43]? What can be gleaned from these rankings?

According to the law in the country where this research was conducted, ethical approval is only required for biomedical research (e.g., MRI, blood sampling, and measuring biological parameters) or research involving underage participants. We have obtained a document from our institution's ethics committee explaining this further (omitted from submission to preserve anonymity).

### 3.1. Study 1

The initial study was meant to be an exploratory analysis of the state of esports training. It was conducted between May 15 and the end

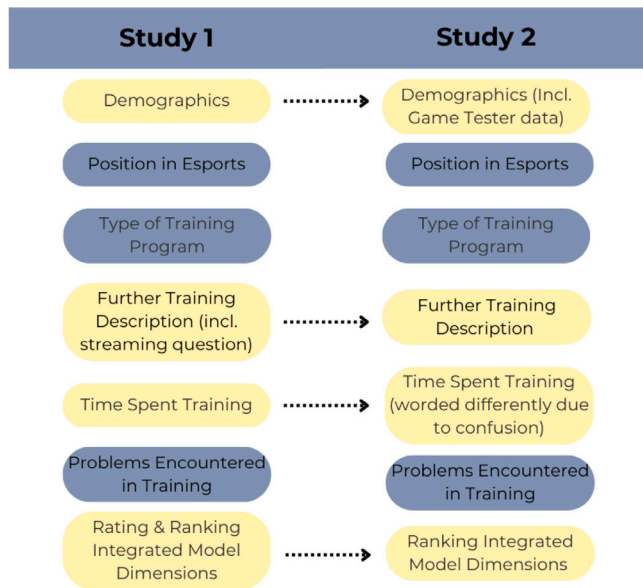


Fig. 1. The comparison between the surveys for Studies 1 and 2, with differences emphasized in yellow. Full list of questions in [Appendices A and B](#).

of August 2022. It included questions about participant demographics, streaming, the *Integrative Model* [43], and current training procedures and the issues with those procedures. We decided that we wanted diverse perspectives, so we tried to target people involved in esports on many levels, not just professional players. This original pilot survey did not go as planned, and we ultimately decided to take the data collected and use it to improve the survey before beginning data collection for the next iteration.

Study 1 consisted of an exploratory survey, which we disseminated widely to get a diverse perspective. Most of the questions were based on demographics or the world of esports.

### 3.1.1. Procedure

The questionnaire had about 28 questions per participant (some had fewer based on their answers) and a consent form. These included demographic questions about age, education level, country of origin, gender, and area of employment. They were also given the option to leave an email address for follow-up studies. Another section asked about their relationship to esports (five questions about streaming, games of interest, and what they did in esports). The final section included experience with/understanding of training. It involved two questions for those not directly involved in esports and 13 for those who were. It was hosted on LimeSurvey [48]. The full list of each question on the survey is in [Appendix A](#).

We planned to recruit participants using a variety of methods in order to get a diverse sample. Our recruitment techniques included social media recruitment (including Reddit, LinkedIn, and X, which was called Twitter at the time of data collection), personal networks (like esports and gaming-focused Discord communities), and cold emailing various esports teams and groups. The goal was to have 30–50 participants from the professional and collegiate field of esports, as well as fan communities. This plan did not succeed, though we were conservative in our goals. We expected to get our target number of participants, even if the numbers would not have reached the heights of prior studies, due to recruiting a specific population and not offering compensation [44,49]. When the data collection period ended, there were 15 complete quality responses. These 15 responses were examined and used to improve the next study.

### 3.1.2. Participants

Fifteen participants participated in our survey; 10 were male, and five were female. The average age was 25.67, and participants' ages ranged from 18 to 42. All but one of our participants finished secondary school or had a GED or some other equivalent. The one participant that did not reported herself as 18, so she possibly had not fully completed her coursework during data collection. The participants reported a variety of countries of origin: three said they were from the United States, two each were from the United Kingdom and Germany, and the rest were from Turkey, Hungary, Slovakia, Romania, Canada, Mexico, Slovenia, and Austria.

### 3.2. Study 2

The prior exploratory study's results were considered as this follow-up study was designed and conducted. Data collection occurred on 21–22 November 2022. The goals of this study were essentially the same as Study 1, just with a larger sample where we paid participants for their time. We wanted to examine the *Integrative Model* [43] as it relates to esports training and participants' perspectives on esports training in general.

The survey consisted of 22 questions, modified from Study 1's 28-question survey. Many questions were reworded, split, combined, or branched out differently to get better data. For example, we reworked the way we phrased the questions related to training hours, and asked more detailed questions to those who did not pursue esports, primarily related to their assumptions.

We eliminated questions about streaming since we wanted to focus on the topic area and decrease the questionnaire's length. However, we did ask about spectating as a form of training, which could involve streaming. We also omitted the Likert ratings of each dimension individually since it would have added excessive length and did not add much to our overall understanding (as [Table 1](#) shows, there was not a wide range in the rankings in the Study 1).

#### 3.2.1. Procedure

This survey was executed via the *Game Tester* platform [3]. The researchers built the questionnaire via *Game Tester's* survey platform, collected data via the same platform, and downloaded the data upon the conclusion of data collection [3]. *Game Tester* also handled the consent forms [3]. There are two sides to the *Game Tester* platform, first the game testing side, where gamers can sign up to be paid for doing surveys like ours, as well as testing games for developers. The other side is the developer/publisher, which pays gamers to test their work [3]. Responses are anonymous because *Game Tester* handles all payment and recruitment. Our institution paid \$376.00 USD to *Game Tester* for distribution. This was after a \$100.00 USD credit was applied for using *Game Tester* the first time. Thus, the total cost was \$476.00 USD. Each participant was paid \$5.95 USD.

As previously mentioned, ethical approval was not required from our institution's ethics committee. However, *Game Tester* did have to approve our survey before it was distributed.

The questions were broken down into three sections: Demographics, Training, and Ranking. There were four demographic questions (age, country, gender, and education level) and up to 13 questions about training. Six of these training questions were multiple choice or quantitative in some way, like "Is/was your training done by following a coach/program?" or "On average, how many hours do/did you spend playing your esports game(s) for training per week?". The remaining seven were more qualitative and included items like "What are/were your primary areas of focus in training?" and "Where do you think your training could be improved?". At the end of the questionnaire was a series of ranking questions. Participants in the Study 1 questionnaire ranked all 19 dimensions in the *Integrative Model*. For this study, we wanted to focus on the most and least important and thus asked participants to choose the three to five most important dimensions and



three to five least important dimensions of the *Integrative Model* [43]. The full list of each question on the survey is in [Appendix B](#).

After downloading the data from *Game Tester*, we ran descriptive statistics for the quantitative measures. We also used texts by Saldana and Mayring to inform qualitative analysis [50–52].

### 3.2.2. Participants

Part of *Game Tester*'s set-up involved selecting what category of players we wanted to target. Depending on their gaming preferences, the players self-select into specific categories on *Game Tester*. These categories are “Mid-core”, “Social”, “Casual”, “Retro”, “Hardcore”, and “Competitive”. We selected the options “Hardcore” and “Competitive” to get experienced players who were more likely to have experience with esports. In fact, of our 40 participants, only three said they played games for under 30 h per week on average.

Forty participants took part in our study; half were in *Game Tester*'s “Competitive” category, and half were in the “Hardcore” category. Three were female, and 37 were male. They were recruited via the *Game Tester* service. The mean age was 26.35. Thirty-four participants had at least some post-secondary education. Of these 40 participants, 25 said they had never competed in or trained for esports, while 15 said they had. Of the 15 who said they were involved in esports, four said they were aspiring pros, one said they were a coach/manager, six said they were on a non-pro team like a school team, and four said they were retired.

Most participants came from Europe or the Americas. Nine came from North America, four from South America, and 22 from Europe (broadly). The remaining five participants came from Bangladesh, the Philippines, the United Arab Emirates, India, and Lebanon.

## 4. Results

We present the results for both Study 1 and Study 2 in this section. However, we first discuss the data analysis process for both studies. We separated our participants into Specialists (who had some relevant experience in the world of esports) and Nonspecialists (who did not have any prior experience in the esports field). This made sense to us as it allowed us to better understand our sample and their perspectives. It also allows us to compare and contrast the two groups because we wanted to see if fan perspectives were wildly different from those who had actually gotten involved with the field.

### 4.1. Data analysis

Due to the low number of participants, we reported descriptive data and did not try to make inferential determinations for either study.

For the qualitative parts of both studies, we used texts by Saldana and Mayring to inform our qualitative analysis [50–52]. Thus, we primarily used combinations of elemental coding methods- like descriptive and concept coding- and content analysis. Participant answers to each question were read multiple times, and we organized their responses into clear thematic categories for each question. For Study 2 in particular, many of our participants gave long, detailed responses, and we attempted to allow for their multi-faceted perspectives, as such, our participants sometimes gave answers that covered multiple thematic categories at once.

### 4.2. Study 1

Despite the fact that we only got data from 15 participants, we still got some interesting results. For example, it seemed that teamwork was one of the most critical overall categories for our participants. This seems reflective of prior findings [45]. Further, most participants who were specialists said that they had little to no formal training resources and that they did not think such a thing existed. This is counter to prior research findings [44].

**Table 1**

Average Likert ratings of dimensions in the integrative model for study 1.

| Dimension                    | Mean  | Std. Dev | 95% CI        |
|------------------------------|-------|----------|---------------|
| Confidence                   | 4.267 | 0.799    | [3.863–4.671] |
| Personal attitudes           | 4.533 | 0.516    | [4.272–4.794] |
| Dealing with pressure        | 4.467 | 0.834    | [4.045–4.889] |
| Motivation                   | 4.133 | 0.916    | [3.669–4.597] |
| Analytical thinking          | 4.067 | 0.961    | [3.581–4.553] |
| Strategic thinking           | 4.333 | 0.900    | [3.878–4.788] |
| Decision making              | 4.867 | 0.352    | [4.689–5.045] |
| Reaction time                | 4.067 | 0.961    | [3.581–4.553] |
| Accuracy                     | 4.133 | 0.743    | [3.757–4.509] |
| Spatial orientation          | 3.800 | 1.014    | [3.287–4.313] |
| Eye–hand coordination        | 4.333 | 0.617    | [4.021–4.645] |
| Teamwork                     | 4.600 | 0.910    | [4.139–5.061] |
| Acceptance of critics        | 4.467 | 0.834    | [4.045–4.889] |
| Coping with technical issues | 3.333 | 1.397    | [2.626–4.040] |
| Adapting to the game setting | 4.067 | 1.163    | [3.478–4.656] |
| Physical strength            | 2.067 | 0.884    | [1.620–2.514] |
| Endurance                    | 3.800 | 0.861    | [3.364–4.236] |
| Speed                        | 3.133 | 0.915    | [2.670–3.596] |
| Agility                      | 3.267 | 1.163    | [2.678–3.856] |

### 4.2.1. Quantitative results

We divide the quantitative results into two primary sections: Training Hours and Integrative Model Dimensions.

**Training hours.** This question did not work out as intended. We asked about overall hours spent training as well as the number of hours spent in several subcategories of training. Our three participants with direct experience in esports reported numbers that lacked logical sense, plausibly because they did not understand the question. The primary issue was that the numbers reported for overall hours spent training and time spent training various subcategories did not add up. We are unsure why this occurred, but it is plausible that when our participants recorded the total hours spent training, they did not account for physical, cognitive, or other types of training. We rephrased and altered our approach to the question for the next iteration. We do not report the data our participants gave because it is clear they did not understand the full scope of these questions and thus did not report good-quality data.

**Integrative model dimensions.** The participants (fans and specialists) rated most items in the *Integrative model* as being of high importance via Likert scales (1 being low importance, 5 being high importance) [43]. “Physical Strength” was the only one of the 19 dimensions with an average rating under 3, while Decision Making had the highest overall ranking at an average of 4.867, as seen in [Table 1](#).

In the forced choice ratings, the dimensions rated within the top three ranks most often by the fans (i.e., were chosen more than three times) were Acceptance of critics (4), Motivation (4), Decision making (6), Personal attitudes (6), and teamwork (8). Over half of the participants cited teamwork as being within the top three most important dimensions of esports training. The other items were primarily personal characteristics rather than training technical skills like reaction time and eye–hand coordination. Only one participant indicated that reaction time was within the top three ranks of training dimensions, while two said eye–hand coordination was within the top three. It is plausible that most fans answered this way because they assumed that technical skills were not something esports professionals needed to focus on due to a pre-existing high skill level.

Our five specialists (including support staff, aspiring professionals, and professionals) somewhat agreed on the most important dimensions with the fans. They selected Personal attitudes (2), Motivation (2), Decision making (2), and Acceptance of critics (2) most frequently. Teamwork only had one vote in the top three, though this could be partially explained by at least one focusing on a single-player esports. This is something reported by Nagorsky and Wiemeyer [43] when they discussed the *Integrative model*. They say that different games and

game types will likely result in different dimensions being more or less important [43].

The three lowest-rank dimensions (selected more than three times) were Agility (10), Coping with technical issues (5), Speed (6), and Physical strength (14). So, only one of the 15 participants did not select “Physical Strength” as low importance. Among the five specialists, Agility (5), Physical strength (5), Speed (4), and Coping with Technical Issues (1) were ranked in the bottom three. There was good agreement between the specialists and fans for the least important dimensions.

#### 4.2.2. Qualitative results

*Series of questions about current training programs.* We asked the participants several questions about their training practices. One of our 15 participants indicated that they were a coach/manager of an esports team, and they indicated that their training program involved “Video replays, analysts, 1 on 1 micro performance, team macro performance, 1 on 1 mindset, strategic coaching”. We also had two aspiring esports professionals. They both said that they were training independently because they were not aware of the existence of any formal training available for their chosen games (*World of Warcraft* [53] and “fighting games”). The two aspiring professional participants said their training focused on improving “mechanical skills” and reviewing their game-play.

*How can training be improved?* We also asked our participants where training could be improved. The participant who said they were a coach/manager said there should be breaks or some “mental reset” between scrimmage and feedback. The independently training participants said they thought they should spend more time training and improving their time management skills. The participants also told us about the weaknesses of their current training procedure. The coach/manager indicated that “Balancing personal life, mental fatigue” was a significant issue. The two aspiring professionals said that the weaknesses of their current training were a lack of equal effort across the team and a lack of structured feedback for improvement.

*Are you streaming? do you think streaming is related to esports training?* Our coach and aspiring professional participants agreed that streaming was an important aspect of esports. The coach reported that streamers often get new players interested in the game itself, while one of the aspiring professionals said that streaming was a good way to record their gameplay.

*Miscellaneous feedback.* At the end of the study, we asked participants if they had anything else to tell us. One of the aspiring professionals said that training “seems poorly optimized”. When we asked the fans if they had anything else to say, they mostly mentioned community and enjoyment. For example, one said “I think the competition, although really cruel, lets the group get emotionally closer together”. Finally, one fan discussed the importance of teamwork, saying “I think in esports teamwork ranks way higher than anything else ... Before and during the game, members have to put trust in their teammates”.

### 4.3. Study 2

The second study was more in-depth and was larger in scale, with forty participants. We break our results into two primary sections: the first reports the quantitative data, while the second discusses the qualitative questions.

#### 4.3.1. Quantitative results

We asked the participants to report which games were related to their esports interests. Many participants listed multiple games; thus, the following numbers do not equal 40 when summed. The participants often gave us specific games, but we report franchises rather than specific games for brevity. Forty-four games total were reported. The most frequently reported game franchises (reported more than ten times each) were the following: *Counter-Strike* (22 participants), *Call of*

*Duty* (20 participants), *League of Legends* (20 participants), *Overwatch* (13 participants), and *Valorant* (10 participants) [54–58]. Readers will observe that these games are played on teams, which is likely to impact Study 2’s results as prior research has found that different types of games require different areas of focus [43,47].

*Training.* Of the 15 participants who said they had some involvement in esports, four indicated that they used coaching or a training program, and the rest were training independently. There were follow-up qualitative questions about the training; these are discussed in the Qualitative Results section.

The 15 individuals who indicated involvement in esports reported an average of 48.467 weekly training hours. They reported an average of 28.867 h of training with their esports game specifically per week, 7.067 h on average doing cognitive training, 5.067 average hours doing physical training, and 7.467 h on average on other types of training. The 25 participants who were not directly involved in esports gave an average estimate within one hour (48.960) to the experts on the average number of hours they felt esports professionals were spending on training.

*Integrative model dimensions.* All participants were asked to select three to five dimensions as “Most Important” and three to five dimensions as “Least Important”. For all 40 participants, the five dimensions most frequently ranked as “Most Important” were “Teamwork” (selected by 85% of participants), “Dealing with Pressure” (chosen by 57% of participants), “Strategic Thinking” (selected by 48% of participants), “Decision Making” (selected by 45% of participants), and “Reaction Time” (selected by 35% of participants). See Fig. 2 for complete details.

The five dimensions selected most frequently by all 40 participants to be “Least Important” were “Coping with Technical Issues” (selected by 63% of participants), “Physical Strength” (selected by 50% of participants), “Agility” (selected by 43% of participants), “Speed” (selected by 35% of participants), and “Personal Attitudes” (chosen by 33% of participants). See Fig. 3 for complete details.

We list the dimensions most frequently chosen as most and least important by group in Table 2. Overall, there was some agreement between the specialists and non-specialists. For the most important dimensions, both groups ranked Teamwork, Dealing with Pressure, and Strategic Thinking as very important dimensions. The Specialists seemed to think that mental status was more important since they ranked dimensions like dealing with pressure and confidence as most important. However, the nonspecialists selected direct game-related skills like Reaction Time as more important. In fact, the specialists said that Personal attitudes and Acceptance of critics were among the most important dimensions, while the nonspecialists frequently chose these same dimensions as least important (see Table 2).

#### 4.3.2. Qualitative results

Not all participants answered every question. The 15 specialists in esports answered the most questions. Interestingly, these forty participants were very willing to share, some writing several paragraphs in response to one prompt (we expected a sentence or two from them per prompt). Thus, their answers often spanned multiple ideas and concepts.

*“Why are/did you train independently? Would you be interested in formal training?”* This question was given to participants who indicated they were involved in esports but were training independently. Eleven participants answered this question.

Five participants indicated an interest in formal training. Participant 9 said “I would be interested in formal training” and Participant 28 said “I would still enjoy receiving formal training, as it would be a great way to test myself and refresh on the best practices for Esports”.

Two of the eleven participants explicitly said they were not interested in formal training. Participant 20 said they are uninterested since they do not see themselves “following the path of Esports”. Participant 26

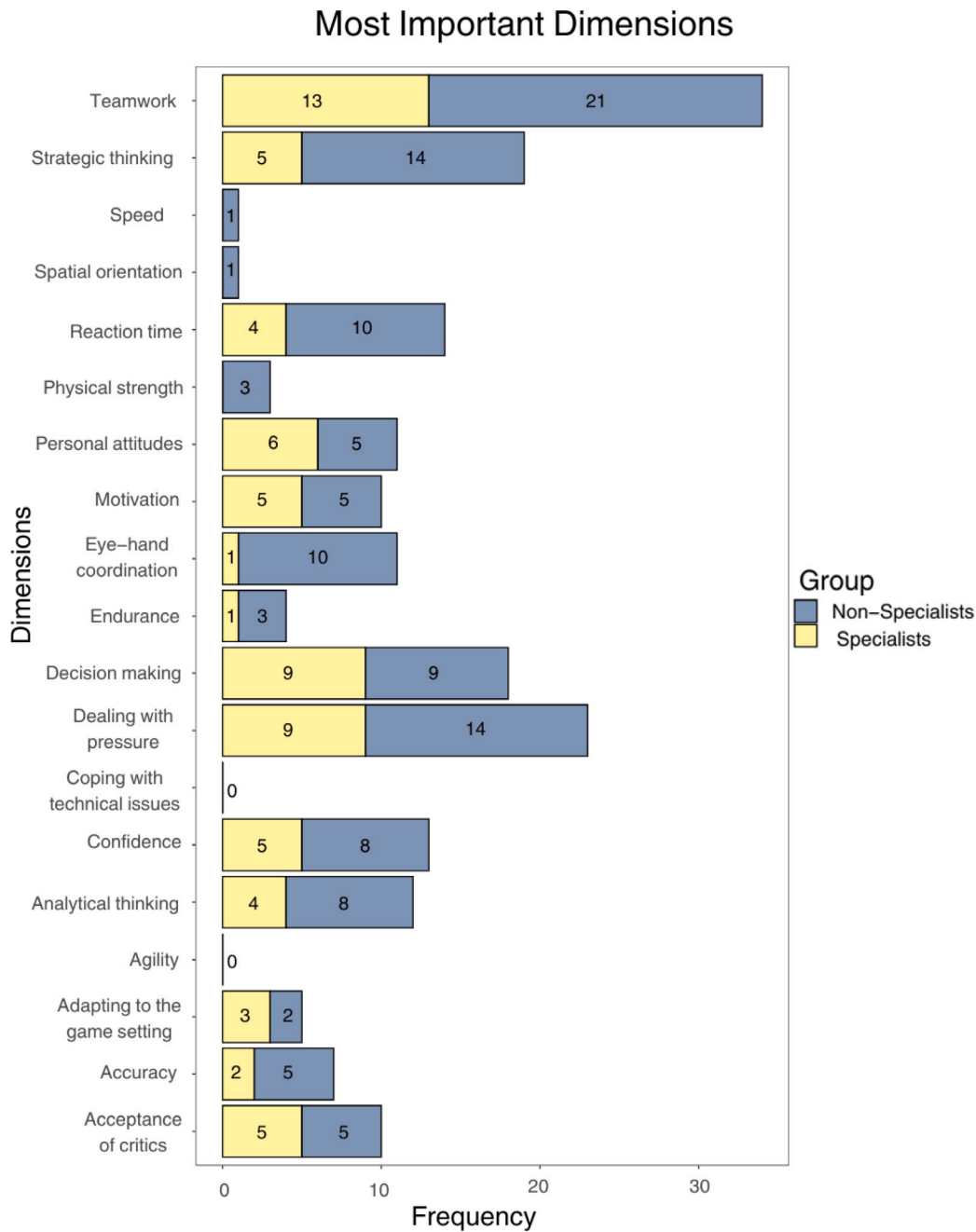


Fig. 2. Integrative model dimensions rated to be most important among 25 non-specialists and 15 specialists in Study 2.

played professionally years ago and moved into coaching themselves. As such, they are not interested in receiving formal training, but they are interested in providing good-quality coaching. The remaining four participants either did not answer whether they were interested or not, or gave neutral responses.

*Reasons for independent training.* Several reasons were given by the eleven participants for why they trained independently. Some, like Participant 12, indicated that formal training was not really around before they retired. Participant 26 said something similar, specifically “a coach was not someone that really existed like we see them today”.

Most of the eleven participants mentioned a lack of opportunity or that their competitions were too small-scale to justify formal training. Participant 17, for example, said “We trained independently because we considered that getting in a coaching program is a waste of money (unaffordable) and the tournament was considered to be in amateur class (regional

tournament)”. Participant 34 also said “I didn’t have the opportunity to meet a pro player or coach. I learn all the tricks, tactics from the internet”.

Another related factor to this lack of opportunity was the lack of finances. Three participants (17, 11, and 26) mention this explicitly, like Participant 17 in the last paragraph. Also, Participant 11 said “We were not as professional or sponsored as other teams and did not have the finances to cover for a coach as most finances was used for travelling”.

“What training program/type of coaching are/were you using?”. This question was given to participants who indicated that they were involved in esports but were training via a coach/program/etc. Four participants answered this question.

Our participants indicated they were using a few different coaching methods; we categorized these as “training programs” and “coaching”. They also mentioned team coordination in cooperative play environments.

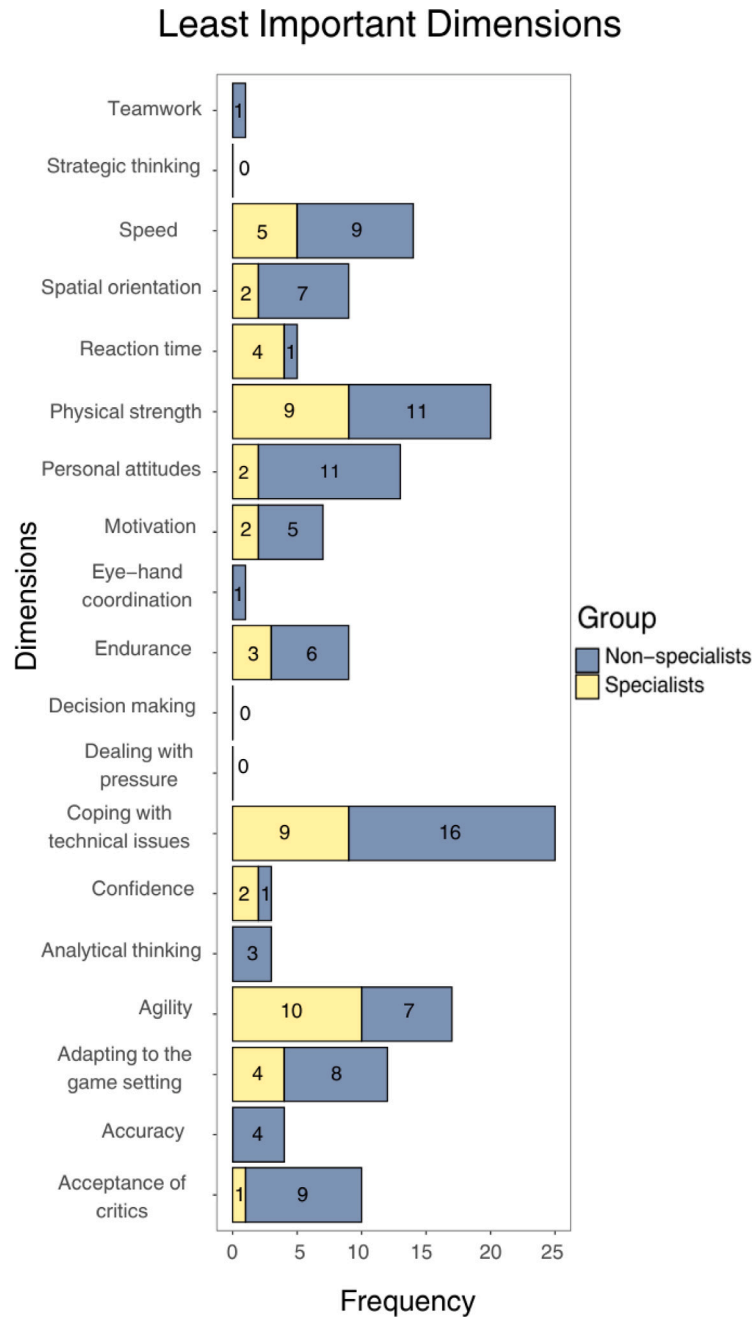


Fig. 3. Integrative model dimensions rated to be least important among 25 non-specialists and 15 specialists in Study 2.

Participant 3 indicated that they “hired a coach from some on-line services” and also mentioned the programs Aimlabs and 3D Aim trainer [59,60]. Participant 13 also mentioned coaching and talked about team coordination. Participant 16 solely discussed team coordination, and Participant 24 reported a program that “mixed a lot of reflex, strategic & tactical thinking, team work and nutrition with some physical training too”. However, they were unable to recall the program’s name.

“Please describe the hardware/equipment/games used for training”. Most of our 15 participants primarily focused on reporting the hardware they had at their disposal, but some also expanded to discuss the software they used.

Our participants primarily used PCs for their esports training, with 12 participants either explicitly stating that they used a PC or alluding to it by discussing their graphics cards and/or peripherals like keyboard, mouse, headset, etc. Two participants mentioned alternatives

to personal computers, with Participant 12 saying they used a PC for one game and a Samsung Smartphone for the other. Participant 20 mentioned the PlayStation 3.

In terms of software, our participants reported many different things. Two participants explicitly reported using the Windows operating system. Aim Trainer and/or Aim Lab was reported by two participants [59,60]. The Rhythm game “Osu!” was also mentioned [61]. According to Participant 1, it helped with reaction times and muscle memory. Participant 3 reported the following games were used in training: *Valorant*, *CS: GO*, *Call of Duty Warzone*, *PUBG PC*, *Super People*, and *Call of Duty Modern Warfare* [54,55,58,62,63].

Participant 13 reported an interesting combination of software, saying that they would play the game during training and used “LoL recording software. We also used Teamspeak to coordinate and skype to attend videocalls were a coach explained strategy using a map of the game and Paint”.



**Table 2**

Five dimensions rated most frequently as most/least important by group in study 2 (Dimensions which were selected an equal number of times are reported together)

|                            | Non-Specialists                            |                       | Specialists                                                         |                       |
|----------------------------|--------------------------------------------|-----------------------|---------------------------------------------------------------------|-----------------------|
|                            | Dimension                                  | Frequency (Out of 25) | Dimension                                                           | Frequency (Out of 15) |
| Most Important Dimensions  | Teamwork                                   | 21                    | Teamwork                                                            | 15                    |
|                            | Dealing with Pressure & Strategic Thinking | 14 each               | Decision Making & Dealing with Pressure                             | 9 each                |
|                            | Reaction Time & Eye-hand Coordination      | 10                    | Personal Attitudes                                                  | 6                     |
|                            |                                            |                       | Confidence, Motivation, Strategic Thinking, & Acceptance of Critics | 5 each                |
|                            |                                            |                       |                                                                     |                       |
| Least Important Dimensions | Coping With Technical Issues               | 16                    | Agility                                                             | 10                    |
|                            | Personal Attitudes & Physical Strength     | 11 each               | Coping with Technical Issues & Physical Strength                    | 9 each                |
|                            | Acceptance of Critics & Speed              | 9 each                | Speed                                                               | 5                     |
|                            |                                            |                       | Reaction Time & Adapting to the game setting                        | 4 each                |
|                            |                                            |                       |                                                                     |                       |

Participant 28 said that they did not use games specifically for training, that they believed that training within their esports game itself was sufficient, they said

“We took advantage of overwatch’s workshop to create some game modes that were more optimized for aim training, team training and tactics training. Our coaches also worked with overwatch’s built-in replay feature to analyze our game-plays, and then report back to the team based on that”.

So it seems that they were using built-in systems within the target game itself, Overwatch [57].

“What problems are encountered during training?”. This question was given to the fifteen participants out of forty who indicated they were “Specialists” (they had been somehow involved in esports at some point). We were able to sort their responses into five categories (it should be noted that some participants gave answers in only one category, while others mentioned items in multiple categories):

- **Knowledge/Skill Issues:** Five participants reported knowledge and skill issues. Participant 11 said keeping up with game tweaks and updates was challenging. Three participants (12, 27, and 39) said they struggled with slow progress and consistency. Participant 39 said “Lack of immediate improvement, hitting a wall, usually is solved by taking breaks and coming back with a fresh mindset”.
- **In-Game Team Communication/Mental Coping Skills:** Another five participants reported difficulties with in-game team communication and mental coping. Participant 1 struggled with processing both in-game events and listening/processing their teammate’s information. They also mention that this leads them to struggle to call out relevant information to their teammates. Other participants reported communication and synergy as well. Participants 13 and 28 discussed keeping the entire team motivated, with Participant 28 saying that it was a challenge to maintain “the team’s morale up even after losing a tournament or something similar”.
- **Team Time Management/Scheduling:** Five participants had problems with time management and scheduling with their team. These scheduling issues were present within the team and made scheduling scrimmages with other esports teams very challenging. Participant 28 sums training problems up with “Trying to coordinate people across multiple timezones, considering all variables and constantly adjusting to make it work as well as possible [is a problem]”.

- **Coaching and Technology Issues:** Although coaching and technology issues were only mentioned by two participants, we felt it important to share what they had to say. Participant 3 said they had “network issues” and sometimes struggled to follow up with their coach’s feedback, although they said this improved over time. Participant 34 was more focused on their reported issues, specifically their technology issues. They said “During training I encountered problems such as: pc crash, internet problems sometimes and power cuts”.

“What are/were your primary areas of focus in training?”. The fifteen Specialists primarily reported that their areas of focus were, unsurprisingly, Personal skills in-game and teamwork. However, two also mentioned another dimension: spectating. We discuss each of these in the following:

- **Personal Skills In-game:** Twelve participants reported personal in-game skills as being a focus in their training. Frequently mentioned items include Aim training, reflexes, strategy, and map awareness.
- **Teamwork:** Eight participants mentioned teamwork as an important dimension of their training. They phrased this in several different ways. Some participants literally listed things like “teamwork”, “coordination”, and “communication” (Participants 9, 16, 24, and 28). Others gave more expansive explanations, like Participant 11, who said “Learning to cope with one another, to trust in your teammates as this really was a team based game” or Participant 20, who said

“Team communication and call-outs were the main focus as we were all well-off mechanically and skill-wise. What separated at our level the good teams from the great teams that made it into LANs was how efficient they communicated and preserved level-heads during less ideal situations”.

- **Spectating:** Two participants explicitly report spectating as a component of training. Participant 5 said that they spectated by watching scrimmages and tournaments. Participant 26, however, endorsed recording and watching their own game-play, “By watching yourself play, you gain awareness in mistakes you continuously make which you can take note of, but also give yourself thought to process to spot other mistakes you might have made, or even things that work well”. Again, it is worth noting that Participant 26 is a retired player and reports that they are currently a coach.

“Where do you think your training could be improved?”. As with other questions, many participants gave multiple answers to this question. We have organized and discussed their responses into the following six categories:

- **Teamwork:** Five participants expressed that they thought their teamwork could be improved. Participant 34 said *“For effective training, it must also include my teammates. They have to coordinate with me and let them know some tactics”*. Participants 11 and 17 said that they felt all teammates should be physically close to each other to encourage team chemistry. This may not be easy, however. Participant 12 lamented that

“I tried to join teams, but I found that in low tiers, people weren’t dedicated enough, and high tiers didn’t want me. So, after some point, I would always leave the teams and keep progressing myself”.

- **Personal Skills In-Game:** Three participants stated that they would like to improve their individual skills, including aim, reflexes, and spatial awareness.
- **Spectating:** Three participants wanted to incorporate more spectating into their training. Participant 9 said that when they wanted their training to be improved, they decided to *“Ask someone to spectate and comment on my gameplay so I don’t become biased and blind to my own failings (Streaming helped a bit with this)”*. Participant 13 also said that they felt they should spectate more frequently. Finally, Participant 20 said that they thought recording and watching their performances would be one way to improve.
- **Physical/Mental Health:** Four participants reported that mental and physical health was something they thought could be improved in training. They mention physical exercise, but also sleep and healthy eating. Participant 9 also says that they think they should incorporate *“other activities besides playing the game”*, perhaps hinting at a desire to have a better work/life balance. Participant 27 agreed, reporting that they *“overtrained”*, and *“as much as you train you need to rest too”*.
- **Guidance/Leadership:** Another five participants said that they wanted better guidance. Participant 13 said their coach lacked experience, whereas the other four said they wished they had any guidance at all. Participant 11 said *“I think if we could afford a coach or trainer it would’ve helped greatly to prepare us for what was to come”*. and participant 16 agreed, saying *“having someone to help coordinate us/ having someone with more experience/ pro esports experience to train us”*, but they added, *“although we had a basic idea of training, we didn’t know how to improve on it”*. So, participants wanted access to coaches who had experience and would be able to guide them on a deeper level.
- **Other:** Two participants also gave answers that did not fit neatly into the five other categories. Participant 24 wanted to improve by upgrading their hardware, and Participant 28 stated they wanted to improve their time management skills.

“Describe your weekly training schedule”. Participants gave a variety of answers here. Some participants said that they did not have a specific weekly schedule (Participant 1), and others said that they had specific routines on the days they did train but did not indicate how many days per week they were training (Participant 23). Others trained in some way almost every day of the week (Participants 9 and 13). Participant 9 said that they would stream for around 6 h and *“My workouts were usually after I finished the training 3 days a week on Monday, Wednesday, and Friday”*. They also said that they tried to train with their entire team against another whole team about once a week (particularly when a tournament was coming up). Participants 16 and 20 reported something similar in that they said that they would scrimmage for a few hours each week.

Three participants said that they trained with their teams after work or class. Participant 17 said that their team of mostly college students played together every evening on weekdays and longer on the weekends. Participants 26 and 34 reported similarly.

Participant 12 took Sundays off to rest, and during the other six days, they reported that they would train for 14 h each day.

Participant 28 described an organized system of shifting timeslots. The total time commitment, they report, was about 15 h (10 on the low end, 20 on the high end).

The lowest amount of training reported came from Participant 24, who reported training twice weekly for two hours.

“What happened when you pursued professional esports?”. Nine of our 15 participants described what happened when they tried to pursue esports. We discuss some of the trends that we saw in this section:

- **Lack of Skill/Time:** One participant, Participant 5, told us that they lacked sufficient skill. Three participants told us that they lacked the time to succeed. Participant 22, for example, reported *“Eventually, I had to prioritize some things over potential Esports career and thus never pursued it after school period”*. Participant 29 reported similarly that their *“Work schedule does not allow for scrimmage”*.
- **Social Support Issues:** Participant 11 reported that they lacked social support and that their family wanted them to pursue financial stability.
- **Team Issues:** Two participants reported team issues. Participant 11 said they could never hone their skills in a team setting because they did not have a fixed team. It is likely that this lack of team is part of the reason they list having no social support in the previous subsection (“Social Support Issues”). Participant 29 was the other participant who listed team issues as a factor, saying *“Age gap, role limitation and roster rotation was conflicting with team”*.
- **Not Giving Up:** On a hopeful note, two participants reported they were still working on their esports goals. Participant 11 said *“I plan on not giving up and keep working on my skills. I still need to make connections and talk to pros and get advices on how to move forward. I might have to work harder but I have decided keep doing my best”*. and Participant 27 said *“so I’m never saying never to new opportunities in life and work hard to achieve new heights”*.
- **Lack of Support in Country:** Three participants said they struggled due to their country’s lack of esports infrastructure. For example, participant 19 reported

“In my country only available tournaments are for games I’m not really interested at, and for games that I adore there are no esports tournaments... I am financially stuck to local competitions”.

Participant 25 answered similarly, saying *“my location is not suitable for Esports development as I am too far from any centers of competition like the ones in Europe or mainland NA”*.

- **Other:** Three participants gave answers that did not fit into any other categories. Participant 4 said nothing had happened to them, although they did not provide any specific reasons. On the other hand, Participant 11 dealt with an illness that impeded their pursuit. They were also streaming, but did not feel that this helped them. Finally, Participant 24 reported that *“I didn’t train enough and didn’t take it serious enough, so it never happened”*. This is unsurprising, given that Participant 24 also reported the lowest number of training hours (approximately four hours per week).

“Why did you not pursue professional esports?”. Sixteen of the participants provided reasons why they never pursued professional esports. Here are the reasons they gave us:

- **Lack of Skill:** Five participants said that they did not have the skills required to pursue esports. Participant 33 said *“I’ve never considered myself a good enough player for ESports level play”*, and Participant 40 agrees, saying that they are *“fully aware that my skills wouldn’t carry me far away in the esports career as im getting older and that decreases my reaction times”*.
- **Lack of Time:** Eight participants reported that they did not have the time to pursue esports. Multiple participants said this lack of time came from other responsibilities, like university in Participant 7’s case or work demands in Participant 30’s case. Other participants said that they did not want to put in the dedication to a single game, like Participant 2, who reported that *“I’m not really a player that could stay in one and only one game, I really love all kind of games and I could say that I can’t move to another one until I finish or get bored from that game”*. Participant 32 also articulated concerns over expending so much energy and time on something that was not guaranteed, saying

“While on paper the idea of earning money by simply being good at video games sounds appealing, I understand that in reality, I would essentially have to dedicate my entire life to improving at a single game in order even to have a small chance of competing professionally and making it a career”.

- **Not Interested:** Five participants reported that they were not interested in pursuing esports in the first place. Participant 2, for example, said that video gaming is a hobby, and they *“don’t want to turn it into something that I could hate”*. Participants 18, 23, and 37 gave similar reasons. Participant 31 said it seemed exhausting, while Participant 37 said they were pursuing another competitive activity.
- **Lack of Support in Country:** Participants 2 and 6 said there was not enough support in their country to pursue esports.
- **Other:** Two participants (8 and 15) gave alternative reasons for not pursuing esports. Participant 8 said they preferred to be a fan, and Participant 15 said they have injuries preventing them from pursuing such a pursuit.

*“In your opinion, what are the areas of training esports professionals are focused on? Are there any other areas they should be focused on in your opinion?”*. Twenty-five non-specialist participants answered this question. Many only answered the first question or stated explicitly that there were no additional areas that esports professionals should be focusing on. Several also said they felt they did not know enough to make recommendations. Nine participants gave complete answers to both questions. All twenty-five participants’ answers were organized into the following five categories:

- **Technical Skill:** Nineteen participants mentioned that technical individual skill sets are a focus for esports professionals. None said that it was an area that esports professionals should focus on more.
- **Strategy and Mental Skill:** Nineteen participants reported that strategy and mental skills are a focus for esports professionals. A few participants indicated that they thought esports professionals should focus on more varied factors related to strategy and mental skill. Participant 15, for example, said that they thought professional esports players were focused on general tactics, but they thought they should practice more *“self-observation”*. Participant 10 advocated for something seemingly similar but referred to it as *“Knowing which style suits you best and work on perfecting it”*.
- **Teamwork and Communication:** Nineteen participants reported that Teamwork and Communication are a focus for esports professionals. Two participants felt that teamwork and communication should receive greater focus. For example, Participant 15 recommended *“Acceptance of critical team feedback”* and Participant 35 thought the professionals *“should focus more on teamwork”*.

- **Physical, Psychological, and Mental Health:** Eight participants believed that Physical, Psychological, and Mental Health were already a focus of esports professionals. Participant 7, for example, reports that esports professionals are already focused on what they call player *“durability”* and stress management. Six participants said that these areas (or particular aspects of these areas) should be focused on more. Participant 14 reports that maintaining physical health is important, while Participant 23 thinks mental health should receive more focus. Participant 22 agrees with both Participants 7 and 23, saying

“I think prioritizing the mental and physical health of an Esports athlete is extremely important which means that there has to be a great work/life balance so that the person doesn’t suffer any illness that can have a severe impact on the person’s life, as well as effect team’s performance as well”.

- **Other:** Two participants gave training recommendations that did not fit well within the other categories. Participant 6 said there should be a greater focus on training by doing. Essentially, increasing player experience by fostering young esports players through tournaments, particularly in countries where esports is still developing. Participant 19 also recommended that esports professionals should practice real-life tasks as training, they report *“In case a game has a mechanic that is also present in real life, it doesn’t hurt to train real life counterpart of that mechanic... For example real life driving experience helped me a great deal with assetto corsa for example, since assetto corsa is a racing simulator”*.

**Other statements of interest.** At the end of the survey, we left space for participants to tell us anything they felt was necessary. In this section, we will report some participant responses that we think are interesting and relevant.

One major trend in these responses involved the lack of infrastructure, which caused problems for esports hopefuls. Participant 9, for example, said

“Where I live I see a lot of young talent for Esports that are not picked up by professional teams because there is little to no incentive from their peers/family and also not enough coaches and professional teams that will really make it possible for them to make a living out of it, so most of them are independent. A lot of players lack motivation because of this money issue and don’t treat it like a real job, so they miss important meetings, matches, etc. The mental health aspect is also very often neglected, and players from big teams crumble from over-training, pressure, and anxiety, so they just give up their careers in ESports”.

Participants 15 and 24 gave similar statements, though Participant 15 also reported that they enjoyed lower-level esports as a teenager.

Finally, participant 32 gave a solid summation of many of the trends we observed:

“In my opinion, many people seem to have a massive misconception about what being a professional player actually entails... in reality most of the work professional players and coaches do is related to the mentality and psyche of the players as well as learning to work in a team environment”.

## 5. Discussion

Throughout these two studies, several factors repeatedly emerged when discussing esports training. We discuss them in conjunction with relevant Research Questions (RQs).

### 5.1. RQ1: Coaching and training programs

In Study 1, the two aspiring professionals said that they were unaware of if there was formal training available for their esports areas.

This differs from prior research, which found that half of the participants used structured training programs and 30% had coaching [44]. We saw different numbers in Study 2's sample, with four out of fifteen (or about 27 percent) of participants indicating that their training was formalized somehow. However, it is worth noting that eight of 15 participants were either aspiring esports players or retired, which may account for the discrepancy.

In Study 2, 11 of the 15 participants who said that they trained independently (without a coach or program) gave three general reasons for training in this way: one, such programs did not exist when they were playing competitively. Second, there was a lack of finances to pay for training, and finally, there was a general lack of opportunity for formal training. Five of them also expressed interest in more formal programs. The four of the fifteen who had formal training seemed to have highly variable experiences. They spoke about coaching feedback, team coordination, and practice with technical skill sets, including programs like Aimlabs and 3D Aim Trainer [59,60]. The primary difference between those who had formal training and those who had informal training would appear to be organization, coordination, and guidance. All participants who were training independently said that they struggled with these areas. However, further research should be conducted in this area as our sample was small.

### 5.2. RQ2: Training done outside of the target esports game

Of the average weekly training hours reported (48.467), almost twenty hours were spent on types of training outside of the chosen esports on average (this included "cognitive", "physical", and "other"). Two participants also explicitly reported that spectating was a primary focus of their training. One participant also said they did not use training tools outside their esports game. There were also frequent mentions of tools like Aim Labs and Aim Trainer [59,60]. These two games are designed to help esports players improve their accuracy and aim in-game. Another interesting inclusion was one participant's report on the use of rhythm games to aid in eye-hand coordination.

### 5.3. RQ3: Places where specialists perceive gaps in training

The most frequently discussed gaps included teamwork, spectating, health (mental and physical), and guidance. In Study 1, our participant, who was a coach/manager, said that mental fatigue was a concern. In study 2, when asked "What problems are encountered during training?" the 15 participants who had experience within esports gave answers that fell into five categories: knowledge/skill issues, in-game team communication/mental coping skills, team time management/scheduling, coaching issues, and technology issues. When we asked where training could be improved, participants gave answers that fit into six categories: Teamwork, Personal Skills In-game, Spectating, Physical/Mental Health, Guidance/Leadership, and the "other" category. As expected, these two questions yielded results that are analogous to each other. These results help us comprehend where specialists perceive the most essential issues researchers and coaches should focus on.

### 5.4. RQ4: Fan and specialists perception of training and where they disagree

The two groups did not universally agree. They, for example, broadly agreed on the importance of teamwork. However, in Study 2 specifically, specialists emphasized mental health while fans were more focused on the technical skillsets, as shown by the forced choice *Integrative model* results. See Figs. 2 and 3 and Table 2 for more details. This discrepancy is likely due to the specialists having a more intimate knowledge of the toll taken by a person in the esports world. Non-specialists may see people "playing video games all day" and miss the substantial pressure placed on players.

### 5.5. RQ5: The Integrative Model

The final research question focuses on Nagorsky and Wiemeyer's [43] *Integrative Model*. We found that several areas seemed overwhelmingly important to everyone: teamwork, mental health, and technical skillsets (see Fig. 2). This is understandable, given that esports professionals need to have the required skills to be competitive but also need to withstand significant pressure. Many of the most popular esports titles right now are also team-oriented games (*League of Legends*, *DOTA 2*, *Valorant*, etc.) [56,58,64], so of course, teamwork is an important facet. Our results reflect these ideas.

In Study 1, with 15 total participants, the following dimensions were rated to be among the most important training dimensions: Acceptance of critics (4), Motivation (4), Decision making (6), Personal attitudes (6), and teamwork (8).

In Study 2, teamwork was selected by 85% of the 40 participants as among the five most important dimensions of training. "Reaction Time", on the other hand, was rated by 35% as being among the most important dimensions. Regarding mental health, "Dealing with Pressure" was selected by 57% of the 40 participants. Also, specialists listed "Personal attitudes", "Confidence", "Motivation", and "Acceptance of Critics" as very important, and each of these dimensions can be associated with mental health. It is interesting, then, that among the esports professionals, teamwork and mental health were often emphasized as something that needed more attention, while technical skill sets were discussed more as a given.

### 5.6. Other observations

We also found some trends that aligned with other aspects of prior research. In terms of time spent training, our results generally agree with previous findings. If we assume that the players in Lee et al.'s [25] sample were training for five to six days each week, our weekly finding of 48.467 h seems comparable to their average of 9.21 per day [25]. Our participants also indicated that they were spending an average of a little over five hours per week on physical training. This aligns more with Kari et al.'s [40] findings that esports players were spending about one hour on physical training each training day, rather than Difrancisco-Donoghue et al.'s [41] finding that 40% of esports players were not getting any physical exercise [40,41]. Also, our findings support prior research that found team cohesiveness to be very important in esports [7,45,46]. We also found that mental health is an issue perceived to be an issue of concern amongst esports specialists, something that has been discussed in other literature as well [1,10,12,23,39,65].

Finally, something we were not explicitly looking for but found interesting was the persistent undercurrent of access or the lack thereof. Many of our participants discussed limitations largely due to a lack of infrastructure in their locations. Team-oriented games are hard to train without living near your teammates (or, at the very least, being within the same timezone). Also, if your game is not well-supported in your country of origin, your chances of getting noticed are low. This lack of access is certainly an area worthy of further study, particularly since one of the advantages of esports is the low barrier of entry [23].

### 5.7. Limitations

There are several clear limitations to this study. The most obvious is that none of our participants were currently working as professional esports players. As such, our conclusions are limited to observations from aspiring pros, coaches/trainers, retired players, and players on nonprofessional teams (like school teams). However, we argue that these perspectives are useful because although these individuals are not professionals, they still interact with the esports world at different levels. Esports training shouldn't only focus on the highest level, the people in the background and the up-and-coming players should be



considered too. We also had a group of participants that was mostly male, which is also a limitation.

Further, we gathered data from 55 participants (if you include Study 1 and 2 combined), and this is not enough people to truly generalize to the entire esports field, thus our conclusions are tentative. Most of our participants came from Europe and the Americas, and we had very little input from esports juggernaut countries in Asia, like South Korea. Further study will be needed to understand the field more holistically. Thus, we now have a general idea regarding answers to our research questions, and these answers can be used to inspire further studies.

As these studies were self-report surveys, we also have limitations relating to design. We cannot verify if our participants were being completely honest. Also, since some of our samples were retired, it is possible that they do not remember everything accurately, even if they are trying to be completely honest. We also do not know if these individuals even are who they say they are, though we attempted to seek out participants in the areas where we thought they might congregate (like games-related forums, hashtags, and the *Game Tester* service). Still, despite these flaws, self-report has long been considered a valuable resource for researchers, provided that we are cautious with our conclusions [66,67]. For example, one paper by Warriner found that self-reports are often inaccurate at the individual level but accurate in aggregation [68]. Thus, we present our results in the context of other work, as this work is primarily meant to be qualitative in nature and thus is more limited in scale.

### 5.8. Future directions

Future research should continue to examine the *Integrative Model* [43]. Perhaps focusing on how dimensions differentiate between game genres. Also, it would be interesting to compare what our sample reports to a practical impact. For example, if we introduce a selection of team-building exercises to an esports team and see how such an intervention impacts their communication and “team cognition” [45].

Further research might also address the following questions: How can we make esports more accessible? How can we foster teamwork for aspiring esports professionals when teammates are not in the same geographic locations? How can we improve teamwork and mental health within the world of esports? How can we help aspiring esports players get structured feedback for improvement? Also, if the results from our relatively small sample are at all repeatable.

Beyond academia, we should also look into how to assist and build up the player bases in countries with little esports-related infrastructure. Existing training programs and opportunities should also maximize their visibility since it seems that many esports hopefuls are unaware of what resources are available to them.

## 6. Conclusion

Ultimately, the esports field is relatively new, and everyone involved is still trying to determine the optimal steps to take next. Data collection from various sources related to esports is an important aspect of forging the path forward. The current study is relatively small-scale; still, the participants we did have were very generous with sharing their perspectives, and this is a qualitative contribution that we think adds nicely to existing literature and can offer a perspective for future study. For example, taking cues from work within traditional sports is likely worth studying. Our findings indicate a need for better mental health and teamwork training resources. Further, we need to ensure that people pursuing esports know what is available, what the best ways to improve are, and how to improve and maintain existing professionals.

### CRediT authorship contribution statement

**Danielle K. Langlois:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Simone Kriglstein:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization.

## Declaration of competing interest

None.

## Appendix A. Study 1 survey questions

1. “What is your age?” (Only inputs above 18 allowed the participant to continue)
2. “Would you be interested in talking to us more about esports via interview? Please input your email if you are (if you are not please leave blank)”
3. “What country do you live in?”
4. “What is your gender?” Options: Male, Female, Nonbinary, Prefer not to answer
5. “What is the highest level of education you have received?” Options: Some secondary school, Secondary school diploma/GED/equivalent, Some post-secondary education, 2-year/associate’s degree, Trade school certification, 4 year/bachelor’s degree, Some graduate school, Master’s degree, PhD or MD or equivalent
6. “Do you have a job outside of Esports?” Options: Yes, No
7. “Briefly describe any employment you have (outside of the Esports area). Please separate different jobs with a semi colon ‘;’” (Only asked if participants answered yes to question 6)
8. “Are you a video game streamer/content creator?” Options: Yes, No
9. “Would you say your streaming/content creation career is related to ESports? If so, how?” (Only asked if participants answered yes to question 8)
10. “What games are related to your Esports job/interest? Please separate different games with a semicolon ‘;’”
11. “Please select the option that most closely aligns with your current position in Esports” Options: Pro (I earn income from esports), New pro (I am signed to a team but have not yet earned income), Aspiring pro (have not yet earned income from esports and I am not signed to a team but practice on a regular basis in order to become a professional), Retired Esports pro (I used to earn income from esports), Coach/manager, Supporting employee of Esports (organizer/host/etc.), Fan, Other (if participants endorsed that they were involved in esports, they answered questions 12 through 26)
12. “On average, how many hours do/did you spend training per week (please include all forms of training)?” (Only asked if participants answered that they were somehow involved in esports in question 11)
13. “On average, how many hours do/did you spend on cognitive/memory/brain training per week?”
14. “On average, how many hours do/did you spend playing your ESports game(s)?”
15. “On average, how many hours of physical training do/did you complete weekly(for example, exercise and stretching)?”
16. “Do/did you engage in other non-physical, non-playing training methods (like watching videos/spectating)?” Options: Yes, No
17. “On average, how many hours do/did you spend on non-physical, non-playing training methods (like watching videos/spectating) per week?” (Only asked if participants answered yes to question 16)
18. “Please select the option that says “violet” to show that you are paying attention”. Options: Red, Blue, Violet, Yellow, Green
19. “Is/was your training done by following a coach/program?” Options: Yes, No
20. “Why are/did you training independently? Would you be interested in formal training?” (Only asked if participants answered no to question 19)

21. "What training program/type of coaching are/were you using?" (Only asked if participants answered yes to question 19)
22. "Please describe the hardware/equipment/games used for training"
23. "What are/were your primary areas of focus in training?"
24. "Describe your training schedule"
25. "Where do you think your training could be improved?"
26. "What problems are encountered during training?"
27. "In your opinion, how important are each of these factors in Esports training? Please rate how important you think each area is for training in Esports (1 being least important, 5 being more important)". Factors: Confidence, Personal attitudes, Dealing with Pressure, Motivation, Analytical thinking, Strategic thinking, Decision making, Reaction time, Accuracy, Spatial orientation, Eye-hand coordination, Teamwork, Acceptance of critics, Coping with technical Issues, Adapting to the game setting, Physical strength, Endurance, Speed, Agility
28. "In your opinion, how important are each of these factors in Esports training? Please rank competency areas you feel to be the most to least important in Esports (drag and drop over to the right box)" Factors: Confidence, Personal attitudes, Dealing with Pressure, Motivation, Analytical thinking, Strategic thinking, Decision making, Reaction time, Accuracy, Spatial orientation, Eye-hand coordination, Teamwork, Acceptance of critics, Coping with technical Issues, Adapting to the game setting, Physical strength, Endurance, Speed, Agility
29. Is there anything else about your experience with Esports and esports training that you would like to tell us?
13. "What are/were your primary areas of focus in training?"
14. "Describe your weekly training schedule"
15. "Where do you think your training could be improved?"
16. "On average, how many hours do/did you spend playing your Esports game(s) for training per week?"
17. "On average, how many hours do/did you spend on cognitive/memory/brain training per week?"
18. "On average, how many hours of physical training do/did you complete weekly (for example, exercise and stretching)?"
19. "On average, how many hours do/did you spend on any other non-physical, non-game-playing training methods (like watching videos/spectating) per week?"
20. "Please select the single option that most closely aligns with your current position in Esports. If multiple of these options apply to you, please select the most relevant to you right now". Options: Support staff/organizer/host, Fan
21. "Have you ever considered pursuing Esports professionally?" Options: Yes, No
22. "What happened when you pursued professional Esports?"
23. "Why did you not pursue professional Esports?"
24. "How many hours, on average, do you think pro Esports players train per week?"
25. "In your opinion, what are the areas of training Esports professionals are focused on? Are there any other areas they should be focused on in your opinion?"
26. "Please select the option that says "violet" to show that you are paying attention". Options: Red, Blue, Violet, Yellow, Green
27. "In your opinion, how important these factors in Esports training? Please select 3 to 5 competency areas you feel are the MOST IMPORTANT in Esports". Options: Confidence, Personal attitudes, Dealing with Pressure, Motivation, Analytical thinking, Strategic thinking, Decision making, Reaction time, Accuracy, Spatial orientation, Eye-hand coordination, Teamwork, Acceptance of critics, Coping with technical Issues, Adapting to the game setting, Physical strength, Endurance, Speed, Agility
28. "In your opinion, how important these factors in Esports training? Please select 3 to 5 competency areas you feel are the LEAST IMPORTANT in Esports" Options: Confidence, Personal attitudes, Dealing with Pressure, Motivation, Analytical thinking, Strategic thinking, Decision making, Reaction time, Accuracy, Spatial orientation, Eye-hand coordination, Teamwork, Acceptance of critics, Coping with technical Issues, Adapting to the game setting, Physical strength, Endurance, Speed, Agility
29. "Is there anything else about your experience with Esports and esports training that you would like to tell us?"

## Appendix B. Study 2 survey questions

1. "What is your age?"
2. "What country do you live in?"
3. "What is your gender?" Options: Male, Female, Nonbinary
4. "What is the highest level of education you have received?" Options: Some secondary school, Secondary school diploma/GED/equivalent, Some post-secondary education, 2-year/associate's degree, Trade school certification, 4 year/bachelor's degree, Some graduate school, Master's degree, PhD or MD or equivalent
5. "What games are related to your Esports job/interest? Please separate different games with a semicolon ','"
6. "Have you ever competed in/trained for Esports (a pro player, aspiring pro, collegiate player, retired pro, coach, etc.)?" Options: Yes, No. I am a fan or support staff (Esports organizer, host, etc.) (If the participant answers yes, the answer questions 7–19, then skip to question 26, if no they answer questions 20–25, then move on to 26.)
7. "Please select the single option that most closely aligns with your current position in Esports. If multiple of these options apply to you, please select the most relevant to you right now". Options: Pro (I earn income from esports), New pro (I am signed to a team but have not yet earned income), Aspiring pro (have not yet earned income from esports and I am not signed to a team but practice on a regular basis in order to become a professional), Retired Esports pro (I used to earn income from esports), Coach/manager
8. "Is/was your training done by following a coach/program?" Options: Yes, No
9. "Why are/did you train independently? Would you be interested in formal training?"
10. "What training program/type of coaching are/were you using?"
11. "Please describe the hardware/equipment/games used for training"
12. "What problems are encountered during training?"
13. "What are/were your primary areas of focus in training?"
14. "Describe your weekly training schedule"
15. "Where do you think your training could be improved?"
16. "On average, how many hours do/did you spend playing your Esports game(s) for training per week?"
17. "On average, how many hours do/did you spend on cognitive/memory/brain training per week?"
18. "On average, how many hours of physical training do/did you complete weekly (for example, exercise and stretching)?"
19. "On average, how many hours do/did you spend on any other non-physical, non-game-playing training methods (like watching videos/spectating) per week?"
20. "Please select the single option that most closely aligns with your current position in Esports. If multiple of these options apply to you, please select the most relevant to you right now". Options: Support staff/organizer/host, Fan
21. "Have you ever considered pursuing Esports professionally?" Options: Yes, No
22. "What happened when you pursued professional Esports?"
23. "Why did you not pursue professional Esports?"
24. "How many hours, on average, do you think pro Esports players train per week?"
25. "In your opinion, what are the areas of training Esports professionals are focused on? Are there any other areas they should be focused on in your opinion?"
26. "Please select the option that says "violet" to show that you are paying attention". Options: Red, Blue, Violet, Yellow, Green
27. "In your opinion, how important these factors in Esports training? Please select 3 to 5 competency areas you feel are the MOST IMPORTANT in Esports". Options: Confidence, Personal attitudes, Dealing with Pressure, Motivation, Analytical thinking, Strategic thinking, Decision making, Reaction time, Accuracy, Spatial orientation, Eye-hand coordination, Teamwork, Acceptance of critics, Coping with technical Issues, Adapting to the game setting, Physical strength, Endurance, Speed, Agility
28. "In your opinion, how important these factors in Esports training? Please select 3 to 5 competency areas you feel are the LEAST IMPORTANT in Esports" Options: Confidence, Personal attitudes, Dealing with Pressure, Motivation, Analytical thinking, Strategic thinking, Decision making, Reaction time, Accuracy, Spatial orientation, Eye-hand coordination, Teamwork, Acceptance of critics, Coping with technical Issues, Adapting to the game setting, Physical strength, Endurance, Speed, Agility
29. "Is there anything else about your experience with Esports and esports training that you would like to tell us?"

## Data availability

The authors do not have permission to share data.

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